	Pimpri Chinchwad Education Trust's **Pimpri Chinchwad College of Engineering & Research Ravet, Pune** An Autonomous Institute | NBA Accredited (4 UG Programs) | NAAC A++ Accredited | An ISO 21001:2018 Certified **IQAC PCCOER**	
AY: 2025– 26	**Assignment No. 4**	Term- I

Department: Computer Engineering **Date:** 22.09.2025
Class: SE **Div:** E&F **Subject Name:** Operating Systems

Course Objectives

Explore the various techniques of memory management.

Course Outcomes

CO No.	Course Outcomes	BT Level
P203.4	Apply techniques of memory management to solve memory management problems	**BT3**

Sr. No.	Question	CO	PI
Q1	Differentiate between internal and external fragmentation	**P203.4** BT4	1.6.1, 2.5.3, 3.5.2, 5.6.2, 9.4.1, 12.4.1
Q2	Explain Page Table structure and Inverted Page Table?	**P203.4** BT2	1.6.1, 2.5.3, 3.5.2, 5.6.2, 9.4.1, 12.4.1
Q3	Explain segmentation with suitable example in brief.	**P203.4** BT2	1.6.1, 2.5.3, 3.5.2, 5.6.2, 9.4.1, 12.4.1
Q4	Given a memory partitions of l00K, 500K, 200K, 300K and 600K (in order), how would each of the first fit, best fit and worst fit algo. Place processes of size 212K, 417K, 112K, 426K (in order)? Which also makes the most efficient use of memory.	**P203.4** BT4	1.6.1, 2.5.3, 3.5.2, 5.6.2, 9.4.1, 12.4.1
Q5	Write proper examples and explain memory allocation strategies first fit, best fit and worst fit. Also explain their advantages and disadvantages	**P203.2** BT2	1.6.1, 2.5.3 3.5.2, 5.6.2 9.4.1, 12.4.1
Q6	Consider page sequence 2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5, 2 and discuss working of following page replacement policies Also count page faults. (use no. of Frames = 3) i) FIFO ii) LRU	**P203.4** BT4	1.6.1, 2.5.3 3.5.2, 5.6.2 9.4.1, 12.4.1

Subject I/C